

A NONLINEAR-APPROXIMATION PERSPECTIVE ON ADAPTIVE DOMAIN DECOMPOSITION FOR PHYSICS-INFORMED NEURAL NETWORKS

ASSAF ZIMAND

ABSTRACT. Physics-Informed Neural Networks (PINNs) provide a flexible framework for solving partial differential equations (PDEs), yet they often struggle with solutions exhibiting spatially heterogeneous complexity. In this work, we introduce an adaptive domain decomposition framework grounded in nonlinear approximation theory. By leveraging connections between decision trees, geometric wavelets, and Besov-type smoothness, the proposed method provides an interpretable and theoretically motivated alternative to black-box adaptive mechanisms. The resulting adaptive mixture-of-experts architecture allocates computational resources according to local function complexity, leading to lighter models with improved or comparable accuracy on benchmark PDEs, including the Schrödinger, Burgers, Allen–Cahn, Korteweg–de Vries, and Kuramoto–Sivashinsky equations.

1. INTRODUCTION

1.1. Motivation. Physics-Informed Neural Networks (PINNs) have emerged as a powerful paradigm for solving forward and inverse problems involving partial differential equations. Despite their flexibility, standard PINNs often struggle when the underlying solution exhibits spatially heterogeneous complexity, such as sharp interfaces, oscillatory behavior, or localized structures. In such settings, a uniform neural network architecture may allocate computational resources inefficiently, often requiring large models to capture localized complex features, which can increase both training time and inference costs.

These challenges motivate the use of adaptive domain decomposition strategies. A principled way to construct such adaptive partitions is provided by nonlinear approximation theory, where hierarchical domain decompositions and tree-based representations have a well-established mathematical foundation. Building on this perspective, we propose an adaptive PINN framework grounded in the theory of nonlinear approximation.

1.2. Related Work. Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. (2019) [14], have emerged as a flexible framework for solving partial differential equations. However, their training is often challenging due to the presence of multiple competing loss terms and the need to compute higher-order derivatives, which can lead to instability and slow convergence [19, 18]. These difficulties have motivated a wide range of improvements to the standard PINN framework.

2020 *Mathematics Subject Classification.* 41A25, 41A46, 65M70, 68T07.

Key words and phrases. Physics-informed neural networks, domain decomposition, nonlinear approximation, Besov spaces, decision trees, mixture of experts.

1.2.1. Domain decomposition methods. A prominent line of work focuses on decomposing the computational domain into subregions. Early approaches include conservative PINNs (cPINNs) by Jagtap et al. (2020) [8], extended PINNs (XPINNs) by Jagtap et al. (2020) [7], and hp-VPINNs by Kharazmi et al. (2021) [9]. More recent methods introduce adaptive decomposition mechanisms, including APINNs by Hu et al. (2022) [6], FBPINNs by Moseley et al. (2021) [13], PINN-Balls by Bonfanti et al. (2025) [1], DIW-PINN by Wang, Y. and others (2025) [20], and Progressive Domain Decomposition (PDD) by Luo et al. (2025) [11]. These methods dynamically allocate subnetworks to regions of high error or complexity, typically relying on residual-based heuristics or learned gating mechanisms.

1.2.2. Architectural improvements. Several works aim to enhance the expressive capacity of PINNs through architectural modifications. The use of random Fourier features by Tancik et al. (2020) [15] mitigates the spectral bias of neural networks and improves their ability to capture high-frequency components. More advanced architectures, such as PirateNet by Wang et al. (2024) [17], modify the standard network structure to improve training stability, especially for deeper models. Additional approaches, such as weight factorization techniques proposed by Wang et al. (2021) [16], aim to improve the network’s ability to represent complex solution patterns.

1.2.3. Adaptive sampling strategies. Adaptive sampling methods refine the distribution of collocation points during training. Residual-based adaptive refinement (RAR) and its variants, including residual-based adaptive sampling (RAD) by Wu et al. (2023) [21], focus computational effort on regions with large residuals. Variational residual-based adaptive approaches, such as vRBA by Hag et al. (2025) [4], further improve sampling efficiency.

1.2.4. Loss balancing techniques. Balancing the different components of the PINN loss function is critical for stable training. Learning Rate Annealing (LRA), introduced by Wang et al. (2021) [19], and related adaptive weighting strategies dynamically adjust the contribution of residual, boundary, and initial condition terms to mitigate gradient pathologies and improve convergence.

1.2.5. Training and optimization strategies. Several works propose modifications to the training procedure itself. Causal training schemes introduced by Wang et al. (2022) [18] enforce temporal structure in time-dependent problems, while time-window approaches by Meng and Karniadakis (2020) [12] decompose long time horizons into smaller subproblems. More recently, optimization strategies have received increasing attention, with studies exploring advanced quasi-Newton methods and optimizer-switching schemes. For example, Kiyani et al. (2025) [10] demonstrated that optimization choices can substantially influence PINN convergence and accuracy. Comprehensive benchmark studies, such as PINNacle by Hao et al. (2024) [5], provide systematic evaluations of these techniques.

While these approaches have demonstrated strong empirical performance, many adaptive decomposition methods rely on trainable mechanisms, such as gating networks or learned weighting functions, to define the interaction between local experts. Although effective, such mechanisms can be difficult to interpret and may behave as black-box components in the decomposition process. In contrast, the present work

proposes a decomposition strategy grounded in nonlinear approximation theory, providing a principled and interpretable framework for allocating computational resources.

1.3. Contributions. The main contributions of this work are as follows:

- **Theory-guided and interpretable adaptive decomposition:** We establish a connection between adaptive domain decomposition in PINNs and nonlinear approximation theory, providing a mathematically grounded framework for constructing hierarchical partitions, while offering improved interpretability and control compared to existing adaptive methods.
- **Adaptive and efficient representations:** The method allocates model capacity according to local solution complexity, enabling efficient representations that use simpler models in smooth regions and higher capacity only where needed.
- **General decomposition framework:** The proposed approach is not tied to a specific PDE formulation and can, in principle, be applied to a broader class of function approximation problems beyond the PINN setting.

1.4. Organization of the Paper. The remainder of the paper is organized as follows. Section 2 introduces the theoretical and mathematical foundations. Section 3 presents the proposed adaptive domain decomposition framework. Section 4 discusses its theoretical interpretation. Numerical experiments are provided in Section 5, followed by discussion and conclusions.

2. BACKGROUND

2.1. PDE Setting and PINN Formulation. We consider a general partial differential equation of the form

$$\mathcal{N}[u](x, t) = 0, \quad (x, t) \in \Omega \times [0, T],$$

subject to appropriate initial and boundary conditions.

In the PINN framework, the solution u is approximated by a neural network u_θ , trained by minimizing a loss function of the form

$$\mathcal{L}(\theta) = \lambda_r \mathcal{L}_r + \lambda_b \mathcal{L}_b + \lambda_i \mathcal{L}_i,$$

where

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{j=1}^{N_r} |\mathcal{N}[u_\theta](x_j, t_j)|^2$$

is the PDE residual loss and the boundary and initial condition losses are given by

$$\mathcal{L}_b = \frac{1}{N_b} \sum_{j=1}^{N_b} |u_\theta(x_j, t_j) - g(x_j, t_j)|^2,$$

$$\mathcal{L}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} |u_\theta(x_j, 0) - u_0(x_j)|^2,$$

where $\{(x_j, t_j)\}_{j=1}^{N_b}$ and $\{x_j\}_{j=1}^{N_i}$ denote collocation points sampled on the boundary and at the initial time, respectively. In practice, these sets are often relatively small compared to the residual points used in $\mathcal{L}_r$.

2.2. Tree-Based Approximation. In contrast to linear approximation methods based on fixed bases or uniform partitions, nonlinear approximation techniques adapt the representation to the local structure of the function. Such methods are known to achieve improved approximation efficiency for functions with spatially varying regularity by allocating resources non-uniformly across the domain (see, e.g., DeVore (1998) [3]).

We consider hierarchical partitions of the domain Ω generated by a decision tree. Each node in the tree corresponds to a region $\Omega \subset \mathbb{R}^d$ and is associated with a piecewise constant approximation

$$Q_\Omega = \frac{1}{|\Omega|} \int_\Omega f(x) dx.$$

Starting from the root region Ω , the domain is recursively partitioned into smaller subregions. Given a region Ω , a split produces a finite collection of child regions $\{\Omega_k\}_{k=1}^m$.

The process continues until a stopping criterion is satisfied, such as a threshold on the approximation error or a minimal number of samples in each region.

2.2.1. Geometric wavelets. Given a parent region Ω_p and a child region Ω_c , the associated geometric wavelet is defined as

$$\psi_c = Q_p - Q_c.$$

The importance of the node is measured by the wavelet norm

$$\|\psi_c\|_2^2 = \|Q_p - Q_c\|_2^2 |\Omega_c|,$$

which quantifies the local contribution of the refinement.

These coefficients are commonly used to guide adaptive refinement, compression, or pruning of the tree representation.

2.3. Besov Spaces and Smoothness. Besov spaces $B_{p,q}^\alpha$ provide a general framework for measuring smoothness that extends classical notions such as Sobolev and Lipschitz spaces, enabling the description of functions with spatially varying regularity.

One characterization of Besov spaces is given in terms of modulus of smoothness. For $f \in L^p$, the modulus of smoothness is defined by

$$\omega_k(f, t)_p = \sup_{\|h\| \leq t} \|\Delta_h^k f\|_{L^p},$$

where Δ_h^k is the k -th order forward difference operator defined by

$$\Delta_h^k f(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x + jh).$$

The Besov norm is given by

$$(1) \quad \|f\|_{B_{p,q}^\alpha} = \|f\|_{L^p} + \left(\int_0^\infty (t^{-\alpha} \omega_k(f, t)_p)^q \frac{dt}{t} \right)^{1/q}$$

Here $\alpha > 0$ denotes the smoothness index, $1 \leq p \leq \infty$ controls the underlying L^p norm, $1 \leq q \leq \infty$, and the integer k is chosen such that $k > \alpha$.

The parameter s quantifies the degree of smoothness of the function, with larger values corresponding to smoother functions. In particular, Besov spaces extend

classical smoothness spaces, recovering Sobolev spaces for $p = q = 2$ and Lipschitz-type spaces for $p = q = \infty$.

2.4. Tree-Based Approximation and Besov Smoothness. The geometric wavelets introduced in the previous subsection induce a notion of smoothness directly on the hierarchical tree structure. This framework, developed in the context of adaptive approximation and geometric wavelets, provides a connection between tree-based representations and classical Besov spaces. In particular, DeVore (1998) [3] established the connection between nonlinear approximation and Besov smoothness, while Dekel and Leviatan (2005) [2] extended these ideas to adaptive tree-based geometric wavelets.

2.4.1. Tree-Besov Norm. In the classical setting, Besov smoothness can be characterized through local moduli of smoothness over dyadic cubes. For

$$\alpha > 0, \quad r \geq 1, \quad \frac{1}{\tau} = \alpha + \frac{1}{p}.$$

the corresponding Besov semi-norm shown in Equation (1) admits the discrete representation

$$|f|_{B_{\tau}^{\alpha,r}} \sim \left(\sum_Q \left(|Q|^{-\alpha} \omega_r(f, Q)_{\tau} \right)^{\tau} \right)^{1/\tau},$$

where the sum is taken over dyadic cubes Q and $\omega_r(f, Q)_{\tau}$ denotes the local modulus of smoothness.

For adaptive tree partitions, an analogous quantity is defined over the regions $\Omega \in \mathcal{T}$:

$$|f|_{B_{\tau}^{\alpha,r}(\mathcal{T})} = \left(\sum_{\Omega \in \mathcal{T}} \left(|\Omega|^{-\alpha} \omega_r(f, \Omega)_{\tau} \right)^{\tau} \right)^{1/\tau}.$$

A fundamental result of Dekel and Leviatan (2005) [2] establishes that the tree-based Besov semi-norm is equivalent to a quantity defined directly from the geometric wavelet coefficients. Specifically,

$$|f|_{B_{\tau}^{\alpha,r}(\mathcal{T})} \sim N_{\tau}(f, \mathcal{T}),$$

where

$$N_{\tau}(f, \mathcal{T}) = \left(\sum_{\Omega \in \mathcal{T}} \|\psi_{\Omega}\|_p^{\tau} \right)^{1/\tau}.$$

2.4.2. Smoothness Index. The corresponding tree-based smoothness index is defined by

$$\alpha^* = \sup \{ \alpha : |f|_{B_{\tau}^{\alpha}(\mathcal{T})} < \infty \}.$$

The quantity α^* measures the smoothness of the function relative to the underlying tree representation. Larger values of α^* correspond to smoother functions.

The equivalence between the Tree-Besov semi-norm and the geometric wavelet norm implies that the decay of geometric wavelet coefficients is directly related to the smoothness of the underlying function. In particular, faster coefficient decay corresponds to higher smoothness. This is analogous to the classical characterization of Besov spaces, where smoothness is reflected by the decay of coefficients in a wavelet

basis. Consequently, geometric wavelet coefficients provide a tree-based characterization of regularity that can be used to quantify local function complexity.

2.4.3. Approximation Rates. The relationship between smoothness and coefficient decay leads to Jackson-type approximation estimates. Let $S_M(f)$ denote the M -term geometric wavelet approximation obtained by retaining the M largest geometric wavelet coefficients. Then, under suitable assumptions,

$$\|f - S_M(f)\|_p \leq CM^{-\alpha} N_\tau(f, \mathcal{T}),$$

where C is independent of M .

This result implies that functions with larger smoothness indices admit more accurate approximations using fewer significant tree nodes. Consequently, geometric wavelet norms and tree-based smoothness measures provide a natural criterion for guiding adaptive refinement and resource allocation.

3. PROPOSED ADAPTIVE DOMAIN DECOMPOSITION METHOD

3.1. Overview of the Method. We propose an adaptive domain decomposition framework in which the computational domain is partitioned into subregions, each associated with a local neural network expert. The decomposition mechanism is based on a decision tree structure, guided by the theoretical connection between tree-based representations and smoothness introduced in Section 2. In contrast, the global composition of local experts follows a partition-of-unity formulation inspired by existing domain decomposition and mixture-of-experts methods.

3.2. Tree Construction and Domain Decomposition. The goal of the decomposition is to partition the domain into regions that reflect the local complexity of the solution. To better understand the desired structure, we construct *reference trees* using ground truth solutions, where the decomposition is determined directly from the geometric wavelet representation. Examples for several benchmark PDEs are shown in Figure 1. These trees provide intuition regarding the regions that are expected to receive additional refinement under an ideal decomposition and serve as a reference for the adaptive procedures considered in this work.

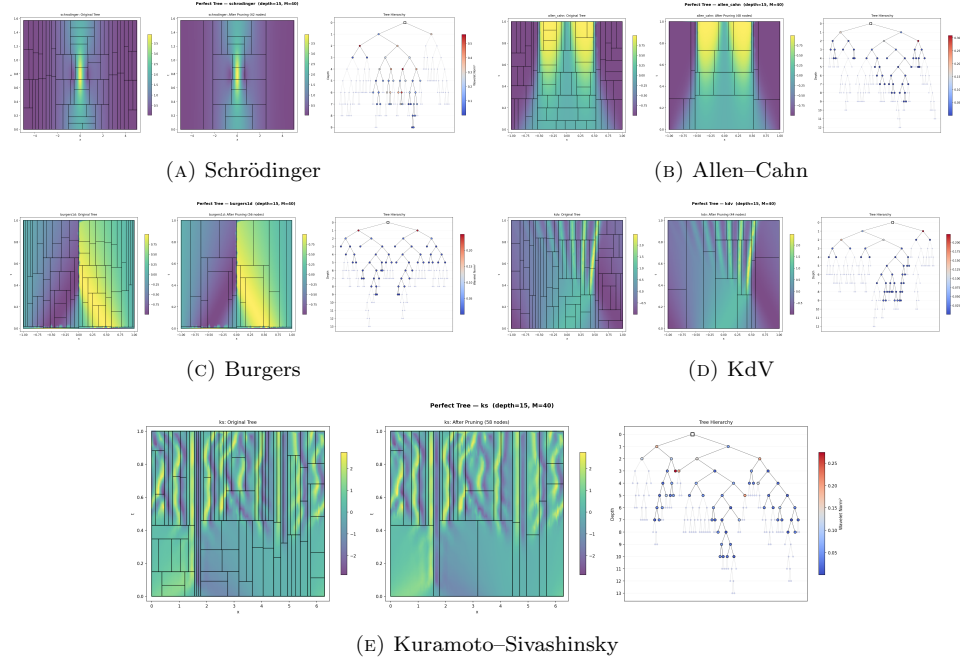


FIGURE 1. Examples of reference trees constructed from ground truth solutions. For each benchmark, the left panel shows the initial tree before pruning, the middle panel shows the retained decomposition after pruning according to the geometric wavelet norm, and the right panel shows the corresponding tree hierarchy. The resulting partitions reveal regions of elevated local complexity and illustrate the structures that the proposed adaptive decomposition aims to recover during training.

[TODO: Update KS to be by window, consider smaller M .]

In practice, we consider two approaches for constructing the tree:

- **Incremental splitting:** training is performed for a fixed number of epochs (or until a convergence criterion), after which a single split is introduced by fitting a shallow decision tree (depth 1). The split is accepted if it satisfies a norm-based criterion motivated by the theory. This approach may be advantageous when an initial vanilla PINN is unable to capture sufficient structure of the solution to reliably estimate a complete tree.
- **M -term tree construction:** a preliminary model is trained using a standard PINN. Although this model is not fully accurate, it often captures the main structure of the solution. A geometric wavelet tree is then constructed from this approximation and compressed using an M -term approximation, retaining the M largest geometric wavelet coefficients. To preserve a valid hierarchical structure, all ancestors of the selected nodes, together with their siblings, are retained, resulting in a complete binary tree.

The M -term construction avoids the need to manually select a wavelet-norm threshold and is generally more robust to such choices. Furthermore, it reduces the interaction between training and tree construction, providing a stable initialization of the decomposition while preserving the most significant structures identified by the geometric wavelet representation. Examples of the resulting decompositions

for several benchmark PDEs are shown in Figure 2, illustrating that the dominant structures of the solution can often be recovered from a relatively short preliminary training stage.

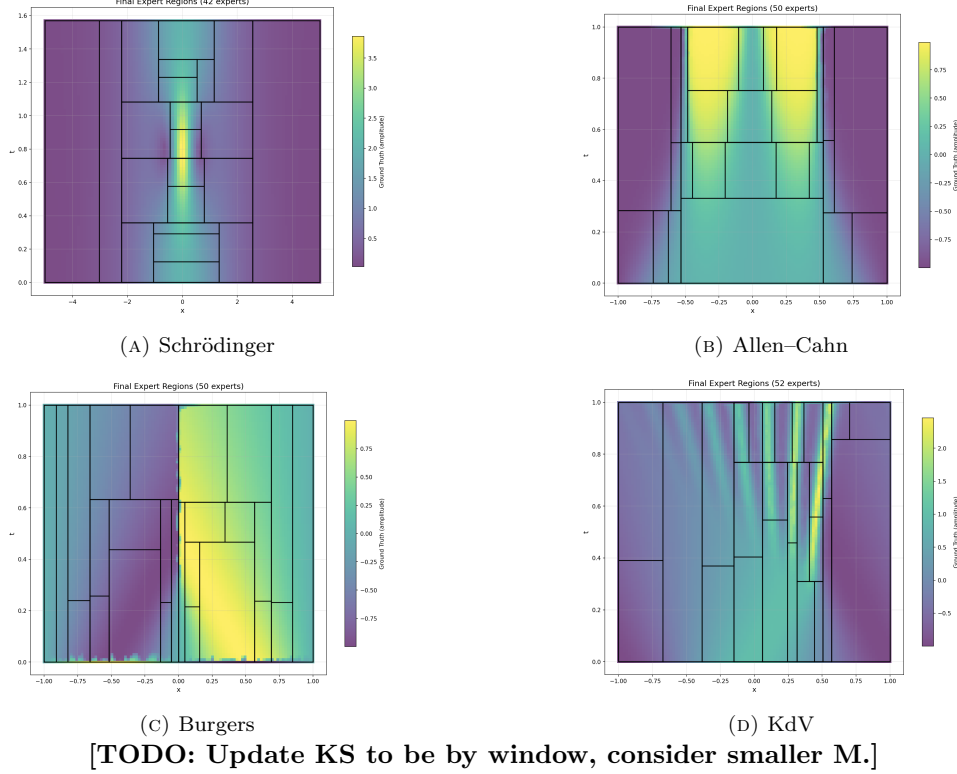


FIGURE 2. Examples of adaptive decompositions obtained using the proposed M -term tree construction procedure. For each benchmark, an initial vanilla PINN was first trained for 2000 epochs using a ResNet architecture. A geometric wavelet tree of maximum depth 15 was then constructed from the resulting approximation and compressed using the M -term-by-norm procedure. To preserve a valid hierarchical structure, all ancestors of selected nodes and their siblings were retained. The resulting trees capture many of the dominant structures observed in the corresponding reference decompositions.

3.3. Local Neural Network Experts. Each subdomain is associated with a neural network expert. We consider standard architectures such as multilayer perceptrons and residual networks.

The capacity assigned to each expert may be adapted according to the local complexity of the corresponding region. In particular, regions associated with larger geometric wavelet norms may require larger models, while smoother regions may be represented using smaller networks. This possibility is motivated by the relationship between geometric wavelet coefficients, smoothness, and approximation rates established in the previous section.

3.4. Global Composition of Local Experts. Let $\Omega \subset \mathbb{R}^d$ denote the computational domain, and let $\{\Omega_i\}_{i=0}^k$ be the set of regions obtained from the hierarchical

decision tree decomposition, where each region is an axis-aligned hypercube

$$\Omega_i = [a_{i,1}, b_{i,1}] \times \cdots \times [a_{i,d}, b_{i,d}].$$

Each region Ω_i is associated with a neural network expert $u_i(x, t)$.

3.4.1. Soft Indicators. To smoothly blend the contributions of different experts, we define a soft indicator function $\Psi_{\Omega_i} : \mathbb{R}^d \rightarrow [0, 1]$ for each region. Let $\sigma(\cdot)$ denote the logistic sigmoid function. For $X = (x_1, \dots, x_d)$, the indicator is defined as

$$(2) \quad \Psi_{\Omega_i}(X) = \prod_{j=1}^d \sigma\left(\frac{x_j - a_{i,j}}{\sigma_{i,j}}\right) \cdot \sigma\left(\frac{b_{i,j} - x_j}{\sigma_{i,j}}\right),$$

where the smoothing parameter is chosen as

$$(3) \quad \sigma_{i,j} = \alpha(b_{i,j} - a_{i,j}),$$

with $\alpha > 0$ controlling the sharpness of the transition near the region boundaries. The expert associated with the root node (corresponding to the entire domain) has an indicator that is constant over the domain, corresponding to a uniform weight before normalization.

3.4.2. Normalization and Partition of Unity. To ensure a proper blending of the experts, the indicators are normalized to form a partition of unity. The normalization depends on the chosen global composition strategy.

- **Normalization over all tree nodes:**

$$(4) \quad \tilde{\Psi}_{\Omega_i}(X) = \frac{\Psi_{\Omega_i}(X)}{\sum_{j=0}^k \Psi_{\Omega_j}(X)}.$$

- **Normalization over leaf nodes only:**

$$(5) \quad \tilde{\Psi}_{\Omega_i}(X) = \frac{\Psi_{\Omega_i}(X)}{\sum_{j \in \mathcal{L}} \Psi_{\Omega_j}(X)},$$

where $\mathcal{L}$ denotes the set of leaf nodes in the decomposition tree. The choice of normalization depends on the strategy used to construct the global solution and on which experts are active during inference. Different variants of the proposed framework combine the contributions of experts in different ways. Some variants use all nodes in the tree, whereas others only activate experts associated with leaf regions. In both cases, the normalized indicators satisfy the partition of unity property

$$\sum_i \tilde{\Psi}_{\Omega_i}(X) = 1.$$

The specific global composition strategies considered in this work are described in the following subsection.

3.4.3. Global Solution Formulations. Using the normalized indicators defined above, the global solution $u_\theta(x, t)$ is constructed from the local experts in three different variants.

- **Adaptive Tree of Experts (AToE).** In this formulation, all experts contribute to the global solution:

$$(6) \quad u_\theta(x, t) = \sum_{i=0}^k \tilde{\Psi}_{\Omega_i}(x, t) u_i(x, t).$$

- **AToE-Leaves.** Only the experts associated with leaf regions participate in the global solution:

$$(7) \quad u_\theta(x, t) = \sum_{j \in \mathcal{L}} \tilde{\Psi}_{\Omega_j}(x, t) u_j(x, t).$$

- **Adaptive Neural Tree (ANT).** In the ANT formulation, information is propagated through the hierarchy by passing the activation of the last hidden layer from each parent expert to its children. Let $A_i(x, t)$ denote the activation of the last hidden layer of expert u_i , and let $\text{parent}(i)$ denote the parent of node i . The output of expert u_i is defined by

$$(8) \quad u_i(A_{\text{parent}(i)}(x, t)) = (u_i^*(x, t), A_i(x, t)),$$

where $u_i^*(x, t)$ denotes the prediction of the expert in the output space of the problem. The global solution is then given by

$$(9) \quad u_\theta(x, t) = \sum_{j \in \mathcal{L}} \tilde{\Psi}_{\Omega_j}(x, t) u_j^*(x, t).$$

All three variants produce a globally defined solution from the same adaptive decomposition and support the allocation of computational resources according to local complexity. The differences between them lie in the interaction between experts and the resulting trade-offs between model capacity, parameter efficiency, information sharing, training stability, and computational complexity.

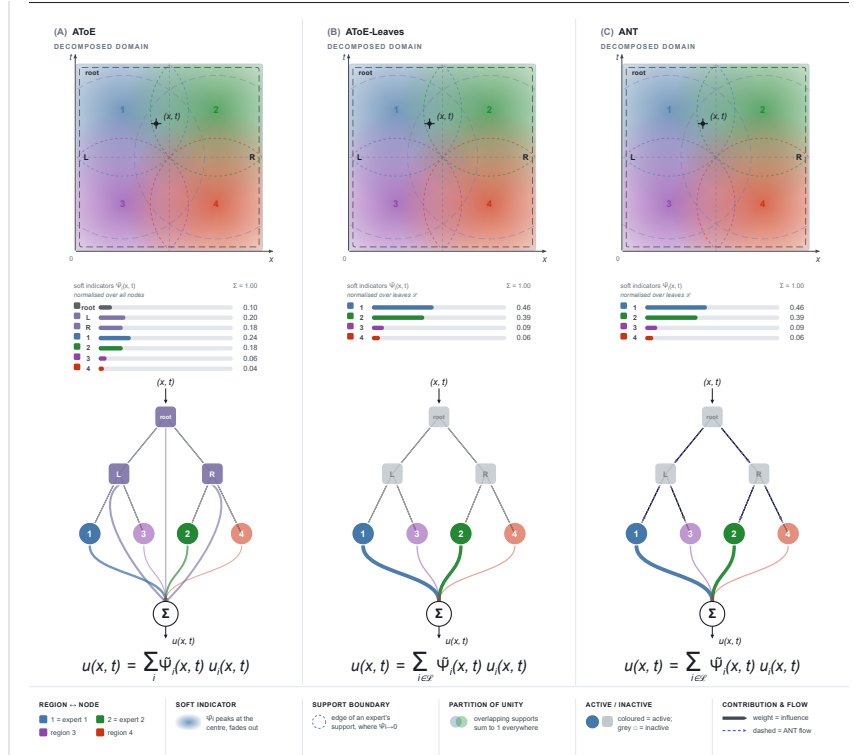


FIGURE 3. Visualization of the AToE, AToE-Leaves, and ANT formulations.

3.5. Training and Implementation Details. The training procedure follows standard PINN optimization, with several enhancements to improve convergence and stability. In particular, we incorporate:

- adaptive sampling strategies,
- causal training for time-dependent problems,
- adaptive loss weighting (e.g., LRA),
- Fourier feature embeddings,
- weight factorization techniques,
- and a second-stage optimizer for refinement.

[TODO: Clarify which techniques are used in final experiments (e.g., PirateNet?).]

4. THEORETICAL INTERPRETATION AND PROPERTIES

4.1. Interpretation of Adaptive Splitting. As discussed in the *Smoothness Index* subsection, the equivalence between the Tree-Besov semi-norm and the geometric wavelet norm implies that the decay of geometric wavelet coefficients is directly related to the smoothness of the underlying function.

A large geometric wavelet norm at a single node does not necessarily imply that the corresponding region is non-smooth, since nodes near the root of the tree may simply capture large-scale variations of the solution. Instead, smoothness is reflected in how rapidly the geometric wavelet norms decay along a branch of the tree. Regions associated with branches whose coefficients remain significant at increasing depths exhibit slower decay and therefore correspond to lower regularity and higher local complexity.

Viewed through this lens, the tree construction procedure described in Section 3.2 naturally favors the retention of branches that exhibit slow coefficient decay across multiple levels of the hierarchy. Consequently, the resulting decomposition concentrates refinement in the less smooth and more complex portions of the domain, while maintaining a coarser representation in regions where the coefficient decay is more rapid.

4.2. Interpretation of Capacity Allocation. The connection between geometric wavelet coefficients and smoothness suggests a second form of adaptivity beyond domain decomposition. In particular, the same theoretical framework can be used to allocate neural network capacity according to local complexity, assigning larger experts to regions associated with slower coefficient decay and smaller experts to smoother regions.

This behavior is analogous to classical approximation settings in which the local approximation order, such as the polynomial degree or the number of retained terms, is adapted according to the smoothness of the target function.

While the experiments presented in this work employ a fixed expert architecture across all subdomains, the same theoretical framework naturally extends to adaptive capacity allocation based on local complexity. This direction is not pursued here and is left for future work.

Nevertheless, even under this restriction, the tree structure itself induces a non-uniform allocation of computational resources. Regions associated with branches that persist to greater depths receive a larger number of experts and therefore a larger effective approximation capacity, whereas smoother regions are represented

by shallower portions of the tree. As illustrated in Figure 4, this mechanism already produces a nonlinear distribution of capacity across the domain that reflects the underlying complexity of the solution.

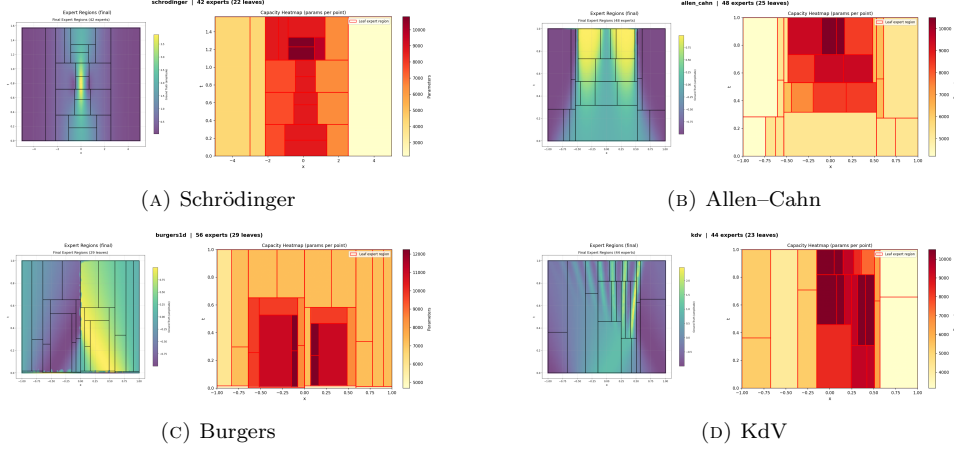


FIGURE 4.

[TODO: Update KS to be by window, consider smaller M, Consider making this per variation with Leaves showing the ratio between volume to capacity]

4.3. Implications for Approximation Behavior. The combination of adaptive partitioning and capacity allocation suggests that the proposed method can achieve improved approximation efficiency compared to uniform models. In particular, the method is expected to provide faster error decay for functions with heterogeneous smoothness by focusing computational resources on regions of high complexity.

[TODO: State informal convergence intuition or reference known rates for adaptive approximation in Besov spaces.]

4.4. Limitations, Scaling, and Open Questions.

4.4.1. Partitioning limits. The effectiveness of the decomposition depends on the ability of the splitting criterion to correctly identify regions of varying complexity. The choice of refinement threshold directly influences the depth of the tree and the resulting distribution of subregions.

4.4.2. Capacity allocation. The assignment of model capacity across regions introduces a trade-off between efficiency and expressivity.

4.4.3. Dimensionality. As the dimension d increases, the number of possible partitions grows rapidly. Understanding how the decomposition behaves in higher dimensions and how it interacts with known limitations of high-dimensional approximation remains an important consideration as the proposed approach is not restricted to PDE settings and can be viewed as a general adaptive approximation framework based on hierarchical partitions and local models, potentially applicable across different domains.

5. NUMERICAL EXPERIMENTS

5.1. Benchmark PDEs. The proposed method is evaluated on five standard benchmark problems commonly used in the PINN literature. Table 1 summarizes the governing equations, computational domains, parameters, initial conditions, and boundary conditions.

PDE	Equation	Domain	Parameters	IC	BC
Burgers	$h_t + hh_x - \frac{\nu}{n} h_{xx} = 0$	$x \in [-1, 1]$ $t \in [0, 1]$	$\nu = 0.01$	$h(x, 0) = -\sin(\pi x)$	$h(-1, t) = h(1, t) = 0$
NLS	$ih_t + \frac{1}{2}h_{xx} + h ^2 h = 0$	$x \in [-5, 5]$ $t \in [0, \pi/2]$	–	$h(x, 0) = 2 \operatorname{sech}(x)$	$h(-5, t) = h(5, t)$ $h_x(-5, t) = h_x(5, t)$
Allen-Cahn	$h_t - Dh_{xx} - 5(h - h^3) = 0$	$x \in [-1, 1]$ $t \in [0, 1]$	$D = 10^{-4}$	$h(x, 0) = x^2 \cos(\pi x)$	$h(-1, t) = h(1, t)$ $h_x(-1, t) = h_x(1, t)$
KdV	$h_t + hh_x + \mu^2 h_{xxx} = 0$	$x \in [-1, 1]$ $t \in [0, 1]$	$\mu = 0.022$	$h(x, 0) = \cos(\pi x)$	$h(-1, t) = h(1, t)$ $h_x(-1, t) = h_x(1, t)$
KS	$h_t + \alpha h h_x + \beta h_{xx} + \gamma h_{xxxx} = 0$	$x \in [0, 2\pi]$ $t \in [0, 1]$	$\alpha = \frac{100}{16}$ $\beta = \frac{100}{16^2}$ $\gamma = \frac{100}{16^4}$	$h(x, 0) = \cos(x)(1 + \sin(x))$	$h(0, t) = h(2\pi, t)$ $h_x(0, t) = h_x(2\pi, t)$

TABLE 1. Benchmark PDEs used in the numerical experiments.

5.2. Experimental Setup. We provide detailed information on neural network architectures, training procedures, sampling strategies, optimizer schedules, and evaluation metrics to ensure reproducibility.

PDE	Optimizer	Scheduler	Epochs	Batch Size	Experts Architecture	Time Windows	Model	Additional Methods	Rel-L2	Ground Truth Comparison
Burgers										
NLS										
Allen-Cahn										
KdV										
KS										

TABLE 2. Experimental configurations and results for the benchmark PDEs.

5.3. Comparison with Baseline Methods. The performance of the proposed approach is compared with standard PINNs and existing domain decomposition methods such as XPINNs and FBPINNs.

5.4. Ablation Studies. We analyze the influence of different design choices, including the three composition variants and the splitting criteria.

5.5. Visualization of Learned Decompositions. The spatial distribution of subdomains is visualized and correlated with regions of high solution complexity.

6. DISCUSSION

6.1. Summary of Findings. We summarize the main theoretical and empirical insights obtained from the proposed adaptive decomposition framework.

6.2. Advantages Over Existing Methods. The interpretability, efficiency, and adaptability of the method are discussed in comparison with existing PINN approaches.

6.3. Limitations. Potential limitations, including computational overhead and sensitivity to hyperparameters, are examined.

6.4. Future Work. Future directions include parallelization of independent experts, extension to higher-dimensional domains, and further theoretical analysis.

7. CONCLUSION

This work introduced an adaptive domain decomposition framework for physics-informed neural networks grounded in nonlinear approximation theory. By connecting hierarchical partitions and local neural network experts with Besov-type smoothness concepts, the proposed method provides an interpretable and efficient approach for solving PDEs with spatially heterogeneous complexity. Numerical experiments on several benchmark problems demonstrate the effectiveness of the approach. Future work will focus on extending the theoretical analysis and improving computational efficiency.

REFERENCES

1. A. Bonfanti, I. Medina, et al., *Pinn balls: Scaling second-order methods for pinns with domain decomposition and adaptive sampling*, arXiv preprint arXiv:2510.21262 (2025).
2. S. Dekel and D. Leviatan, *Adaptive multivariate approximation using binary space partitions and geometric wavelets*, SIAM Journal on Numerical Analysis **43** (2005), no. 2, 707–732.
3. Ronald A. DeVore, *Nonlinear approximation*, Acta Numerica **7** (1998), 51–150.
4. J. Hag et al., *A variational framework for residual-based adaptivity in neural pde solvers*, npj Artificial Intelligence (2025).
5. Z. Hao et al., *Pinnacle: A comprehensive benchmark of physics-informed neural networks*, NeurIPS (2024).
6. Z. Hu, A. D. Jagtap, and G. E. Karniadakis, *Augmented physics-informed neural networks (apinns): A gating network-based soft domain decomposition methodology*, arXiv preprint arXiv:2211.08939 (2022).
7. A. D. Jagtap, K. Kawaguchi, and G. E. Karniadakis, *Extended physics-informed neural networks (xpinns): A generalized space-time domain decomposition framework*, Communications in Computational Physics **28** (2020), no. 5, 2002–2041.
8. A. D. Jagtap, E. Kharazmi, and G. E. Karniadakis, *Conservative physics-informed neural networks on discrete domains*, Computer Methods in Applied Mechanics and Engineering **365** (2020), 113028.
9. E. Kharazmi, Z. Zhang, and G. E. Karniadakis, *hp-vpinns: Variational physics-informed neural networks with domain decomposition*, Computer Methods in Applied Mechanics and Engineering **374** (2021), 113547.
10. Shayan Kiyani, Roberto Molinaro, and Marta D’Elia, *Comparing optimization strategies for physics-informed neural networks and physics-informed kolmogorov–arnold networks*, arXiv preprint arXiv:2501.16371 (2025).
11. D. Luo et al., *Progressive domain decomposition for efficient training of physics-informed neural networks*, Mathematics **13** (2025), no. 9.
12. X. Meng and G. E. Karniadakis, *A composite neural network that learns from multi-fidelity data: Application to function approximation and inverse pde problems*, Journal of Computational Physics **401** (2020).
13. B. Moseley, A. Markham, and T. Nissen-Meyer, *Finite basis physics-informed neural networks (fbpinns): A scalable domain decomposition approach*, arXiv preprint arXiv:2107.07871 (2021).
14. M. Raissi, P. Perdikaris, and G. E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics **378** (2019), 686–707.
15. M. Tancik, P. Srinivasan, et al., *Fourier features let networks learn high frequency functions in low dimensional domains*, NeurIPS (2020).

16. S. Wang et al., *On the eigenvector bias of fourier feature networks: From regression to solving multi-scale pdes*, Computer Methods in Applied Mechanics and Engineering (2021).
17. S. Wang, B. Li, Y. Chen, and P. Perdikaris, *Piratenets: Physics-informed deep learning with residual adaptive networks*, Journal of Machine Learning Research **25** (2024).
18. S. Wang, S. Sankaran, and P. Perdikaris, *Respecting causality in physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering (2022).
19. S. Wang, Y. Teng, and P. Perdikaris, *Understanding and mitigating gradient pathologies in physics-informed neural networks*, SIAM Journal on Scientific Computing **43** (2021), no. 5.
20. Y. Wang et al., *Diw-pinn: Domain decomposition based on wavelet analysis for physics-informed neural networks*, Nonlinear Dynamics (2025).
21. C. Wu, M. Zhu, et al., *A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks*, Computer Methods in Applied Mechanics and Engineering **403** (2023), 115671.

Email address: `assafzimand@mail.tau.ac.il`